MTN Uganda Escrow Bank Integration document

MTN Uganda

Rev PA1

Copyright

Contributors

Name	Role
Agaba Simon Peter	Senior Business Analyst

Revision History

Revision	Date	Comments
PA1	03-04-2019	Initial Draft

Contents

1 Introduction

1.1 Requirement

This solution document is for a bank that wants to preform escrow integration into MTN. Escrow integration comprises of two sets of use cases;

1) Money deposits and withdrawals
2) Reconciliations and compliance.

MTNs expectation is that deposits are exclusively to Agents (if the profile contains the word agent from the getaccountholderinfo) and they use MSISDNs of format 07XX but are sent in international format 2567xxx and to service providers who use External IDs that begin with 5xxxx, for details on how to send the different types refer to the FRI Types (Section 2.1.9)

1.2 Scope

Design includes:

- Advise on APIs to be used and all APIs are based on XML/HTTP
- Design on flows
- Integration guidelines, no TLS protocol less than TLS1.2 will be allowed

1.3 Intended Audience

The following are included in the target group for this document:

- Solution Architects
- Account Managers
- Project Managers
- System Integrators

2 Implementation flow

2.1 API Integration flow

The implementation is such that the partner bank will develop both the deposit and withdrawal APIs since for escrow integrations authentication is directly with the core mobile money platform. MTN uses a single BIN approach with it's customers implying that all customers in the mobile money platform don't belong to any bank and deposits and withdrawals can be effected from any escrow partner. We have the following APIs; The mobile money platform identifies users using an FRI (Financial resource Information) in this case all MSISDNS must be sent as FRI:256772121243/MSISDN, International format is a MUST.

2.1.1 Getaccountholderpersonalinfo

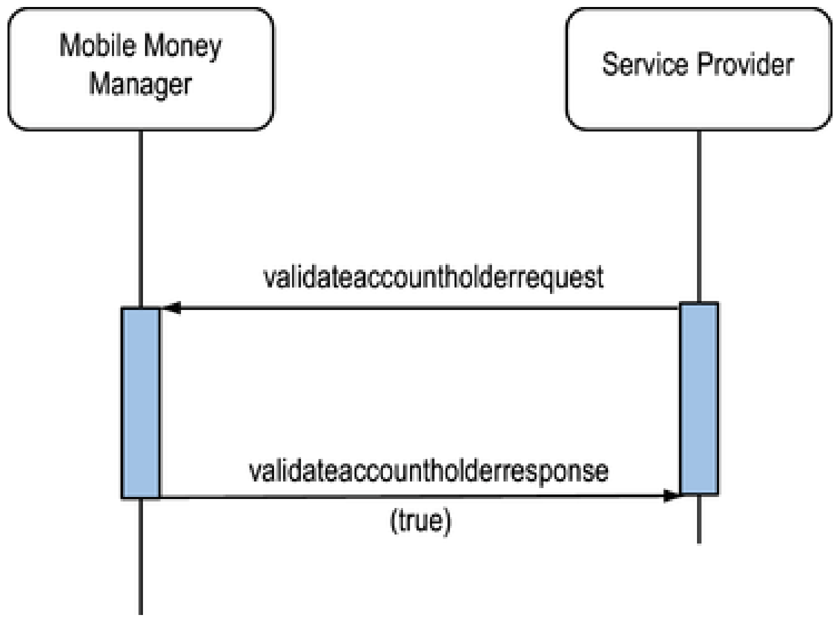
The getaccountholderpersonalinfo API is used to query the names of an account holder for validation purposes. This API isn't mandatory but is good for customer experience.

```
<ns0:getaccountholderpersonalinformationrequest
xmlns:ns0="http://www.ericsson.com/em/emm/provisioning/v1_0">
  <identity>ID:256789608735/MSISDN</identity>
</ns0:getaccountholderpersonalinformationrequest>
```

```
<ns4:getaccountholderpersonalinformationresponse
xmlns:ns4="http://www.ericsson.com/em/emm/provisioning/v1_0"
xmlns:prc="http://www.ericsson.com/em/emm/provisi
oning/v1_0/common" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <information>
    <name>
      <prefix>DOCT</prefix>
      <firstname>Jane</firstname>
      <surname>Doe</surname>
    </name>
    <birth>
      <date>1988-10-01</date>
      <country>UG</country>
    </birth>
    <occupation/>
    <residentialstatus/>
  </information>
</ns4:getaccountholderpersonalinformationresponse>
```

2.1.2 Validateaccountholder

This is a validation API that is used to confirm if a user can accept or not accept payments. This is an optional API but in the case of real time transactions over the counter it can help together with the getaccountholderpersonalinfo validate a subscriber.

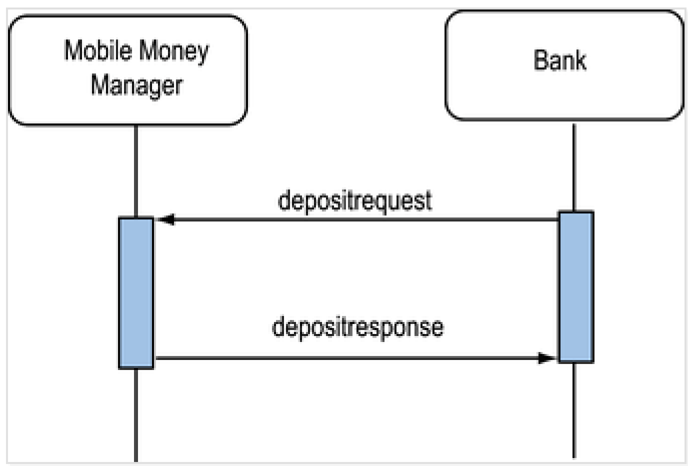


<ns0: validateaccountholderrequest xmlns:ns0="
http://www.ericsson.com/em/emm/serviceprovider/v1_0/backend ">
<accountholderid>ID:256789608735/MSISDN</accountholderid>
</ns0:validateaccountholderrequest>

<ns0: validateaccountholderresponse xmlns:ns0="
http://www.ericsson.com/em/emm/serviceprovider/v1_0/backend ">
< valid >TRUE</valid>
</ns0:validateaccountholderresponset>

2.1.3 Deposit api

The deposit API is used to perform deposits into a Mobile Money account, the assumption is that money has already been deposited in the escrow as a result in the case of none existent or fault MSISDNs, manual resolution must take place at the MTN end. This is a Mandatory API for deposits. query the names of an account holder for validation purposes. This API supports duplicate ID detection and as part of testing duplicate transactions should be tested to ensure expected handling, no deposit will be processed twice for the same transaction ID. Deposits with duplicate transaction ID will fail and return error code SETTLEMENT_DUPLICATE_RECORD_FOUND.

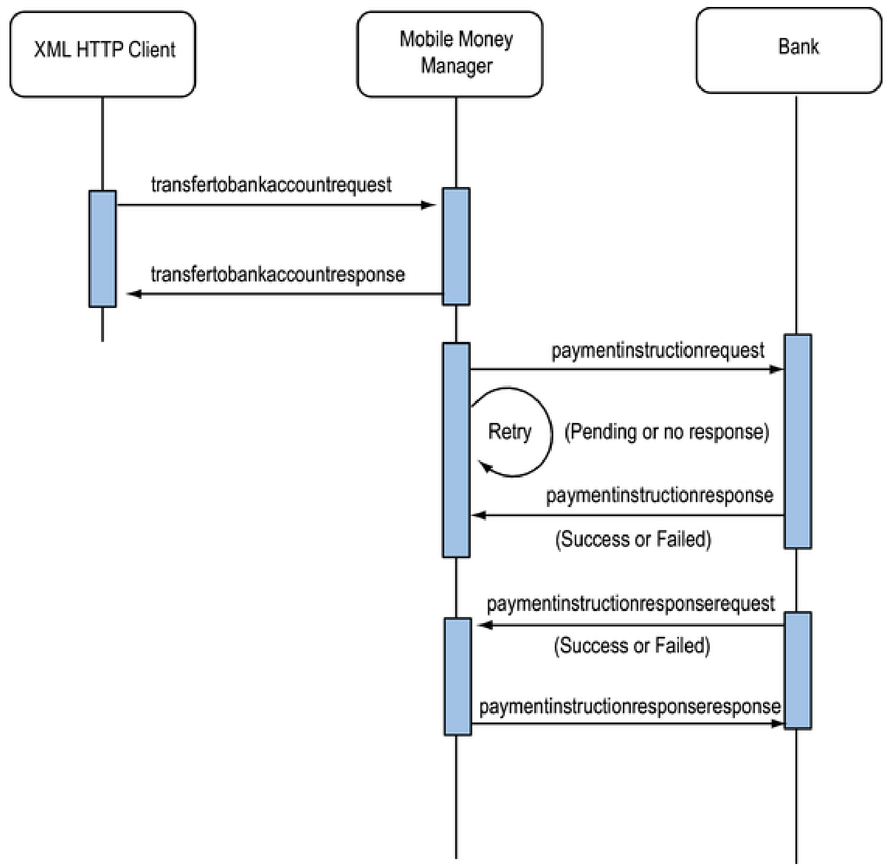



```
<ns0:depositrequest xmlns:ns2="http://www.ericsson.com/em/emm/settlement/v1_0">
   <bankcode>10001</bankcode>
   <accountnumber>77020550413</accountnumber>
       <transactiontimestamp>
      <timestamp>2017-08-22T01:00:00.000+03:00</timestamp>
   </transactiontimestamp>
   <amount>
      <amount>500</amount>
      <currency>UGX</currency>
   </amount>
   <receiver>FRI:256772712561/MSISDN</receiver>
   <banktransactionid>45271123</banktransactionid>
   <receiverfirstname>Test</receiverfirstname>
   <receiversurname>Test</receiversurname>
   <message>My Float</message>
</ns0:depositrequest>

<ns0:depositresponse xmlns:ns4="http://www.ericsson.com/em/emm/settlement/v1_0">
<status>SUCCESS</status>
<financialtransactionid>2571948</financialtransactionid>
</ns0:depositresponse>
```

2.1.4 paymentinstruction

The paymentinstruction API is used to trigger liquidations from the mobile money platform to the bank. This API is **mandatory** for liquidations and will send the same Transaction ID for the same withdrawal request. The bank must be able to support unique transaction IDs to ensure no duplicate withdrawals are effected. This API is synchronous in nature. This API supports retries until success or failed responses are sent and has a timeout value of 20 seconds.




```
<ns2: paymentinstructionrequest  xmlns:ns2="http://www.ericsson.com/em/emm/settlement/v2_0">
  <transactiontimestamp>
    <timestamp>2001-12-31T12:00:00</timestamp>
  </transactiontimestamp>
  <amount>
    <amount>50.0</amount>
    <currency>UGX</currency>
  </amount>
  <paymentinstructionid>12345</paymentinstructionid>
  <receiverbankcode>CERUBUGUXXX</receiverbankcode>
  <receiveraccountnumber>1231414141441</receiveraccountnumber>
  <receiverfirstname>John</receiverfirstname>
  <receiversurname>Doe</receiversurname>
  <transactionid>4619893691</transactionid>
  <message>message</message>
</ns2:paymentinstructionrequest>
```

```
<ns2:paymentinstructionresponse xmlns:ns2="http://www.ericsson.com/em/emm/settlement/v2_0">
<status>SUCCESS</status>
  <paymentinstructionid>12345 </paymentinstructionid>
  <banktransactionid>12345</banktransactionid>
  <transactiontimestamp>
    <timestamp>2001-12-31T12:00:00</timestamp>
  </transactiontimestamp>
  <bookingtimestamp>
    <timestamp>2001-12-31T12:00:00</timestamp>
  </bookingtimestamp>
  <amount>
    <amount>500.0</amount>
    <currency>UGX</currency>
  </amount>
</ns2:paymentinstructionresponser>
```


2.1.5 getcustodyaccounttransactionhistory

The getcustodyaccounttransactionhistory API is used to query custody accounts transaction history and view the deposits and withdrawals over a given period.


```
<ns2:getcustodyaccounttransactionhistoryrequest
xmlns:ns2="http://www.ericsson.com/em/emm/financial/v1_3">
<bankcode>1234</bankcode>
<accountnumber>0123456789</accountnumber>
<startdatefrom>2019-03-05T00:00:00Z</startdatefrom>
<startdateto>2013-09-05T23:00:00Z</startdateto>
</ns2:getcustodyaccounttransactionhistoryrequest>
```


** Alternatively fnumoftransactions can be used with a manageable number of 50 and an indexoffset to step through different sets of data

Sample response


```
<?xml version="1.0" encoding="UTF-8"?>
<ns6:getcustodyaccounttransactionhistoryresponse
xmlns:ns6="http://www.ericsson.com/em/emm/financial/v1_3"
xmlns:fic="http://www.ericsson.com/em/emm/financial/v1_3/common"
xmlns:ns2="http://www.ericsson.com/em/emm/v1_0/common"
xmlns:op="http://www.ericsson.com/em/emm/v2_0/common"
xmlns:xs="http://www.w3.org/2001/XMLSchema">
   <totalcount>1</totalcount>
   <historylist>
      <entry>
         <financialtransactionid>2570713</financialtransactionid>
         <transactionstatus>SUCCESSFUL</transactionstatus>
         <transfertype>DEPOSIT</transfertype>
         <startdate>2017-04-11T11:44:04.102+03:00</startdate>
         <commitdate>2017-04-11T11:44:04.566+03:00</commitdate>
         <initiatinguser>ID:centbank/ADMIN</initiatinguser>
         <realuser>ID:centbank/ADMIN</realuser>
         <direction>OUT</direction>
         <from>FRI:1508144/MM</from>
         <fromaccount>FRI:1508144/MM</fromaccount>
         <fromamount>
            <amount>2000.00</amount>
            <currency>UGX</currency>
         </fromamount>
```

```
      <originalamount>
        <amount>2000.00</amount>
        <currency>UGX</currency>
      </originalamount>
      <to>FRI:256779999508/MSISDN</to>
      <toaccount>FRI:1498181/MM</toaccount>
      <toamount>
        <amount>2000.00</amount>
        <currency>UGX</currency>
      </toamount>
      <tomessage>My Float</tomessage>
      <maininstructionid>10100</maininstructionid>
      <instructionid>10023</instructionid>
      <externaltransactionid>3512172222</externaltransactionid>
      <toaccountholder>ID:440728/ID</toaccountholder>
      <banktransactionid>3512172222</banktransactionid>
      <banktransactiontype>DEPOSIT</banktransactiontype>
    </entry>
  </historylist>
</ns6:getcustodyaccounttransactionhistoryresponse>
```


2.1.6 getaccountholderinfo: API Used to validate Agents

This API is used to query the user details, it returns details like the names;


```
<ns0:getaccountholderinforequest
xmlns:ns0="http://www.ericsson.com/em/emm/provisioning/v1_2">
  <identity>ID:256772712716/MSISDN</identity>
</ns0:getaccountholderinforequest>

2017-05-26 00:33:59.068 TRACE http-78 c.e.em.am.http.client.HttpClientImpl -
<?xml version="1.0" encoding="UTF-8"?>
<ns5:getaccountholderinforesponse
xmlns:ns5="http://www.ericsson.com/em/emm/provisioning/v1_2"
xmlns:ns3="http://www.ericsson.com/em/emm/v1_2/common"
xmlns:op="http://www.ericsson.com/em/emm/v1_0/common"
xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <accountholderbasicinfo>
    <msisdn>256772712716</msisdn>
    <firstname>Created</firstname>
    <surname>ByScript</surname>
    <profilename>MTNU Mobile Money Agent Bronze</profilename>
    <internalidentity>ID:470780/ID</internalidentity>
    <defaultfris>
      <defaultfri>
        <fri>FRI:1528233/MM</fri>
        <currency>
          <code>UGX</code>
        </currency>
      </defaultfri>
    </defaultfris>
    <loyaltypointsaccountfri>FRI:1528232/MM</loyaltypointsaccountfri>
    <acceptedtc/>
    <accountholderstatus>ACTIVE</accountholderstatus>
    <bankdomainname>MTNUGANDA</bankdomainname>
    <hasparent>false</hasparent>
    <languagecode>
      <code>en</code>
```

</languagecode>
<children xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:nil="true"/>
</accountholderbasicinfo>
</ns5:getaccountholderinforesponse>

2.1.7 Error response xml API

Below is the sample error response XML.

<ns0:errorResponse xmlns:ns0="http://www.ericsson.com/lwac" errorcode="COMMUNICATION_ERROR"></ns0:errorResponse>

<ns2:errorResponse xmlns:ns2="http://www.ericsson.com/lwac" errorcode="TRANSACTION_TIMED_OUT"/>

Error codes can vary based on actions by a subscriber or their account or the 3rd party transaction status, hence we have codes like COMMUNICATION_ERROR or VALIDATION_ERROR

2.1.8 Error Handling

When an operation results in an operation error, the HTTP connection will respond with the code 500. The operation errors may for example be invalid permissions, invalid parameter values or internal errors of the system. If response code 500 is returned, it is possible to unmarshall the errorresponsetype object from the error stream of the connection.

The actual error codes are subject to change so they are currently not included in the description of each operation. Generally, the text for an error code allows a person to interpret the error correctly. For example, the code ACCOUNTHOLDER_NOT_FOUND indicates that an identity was provided that does not have a match in the system.

The error code RESOURCE_NOT_FOUND similarly indicates that the requested financial resource, such as an account in the specified currency, does not exist in the system.

Note:

Prevent leaking too much information to an end user of the system. For example, when requesting a voucher that does not exist, the error VOUCHER_NOT_FOUND is returned. When requesting a voucher that exists but with another owner, the error PERMISSION_DENIED is returned. These two errors must not be displayed to an end user since they expose a possibility to use brute force to get hold of valid voucher tokens in the system.

Below is the table for error handling

errorResponse

Name	Type	Multiplicity	Description
errorcode	xs:string	Mandatory	Refer to future list of error codes.
arguments	tns:errorargumenttype	Optional, 0..N	See `errorArguments`.

errorArgument				
	name	xs:string	Mandatory	Additional info depending on error code.
	value	xs:string	Mandatory	Additional value depending on error code.

2.1.9 Financial Resource Identifier

The FRI is an identifier, pointing to one or more financial resources, such as accounts, vouchers, or other financial resources. The FRI is, together with the transaction currency, used to identify, for a particular transaction, the sending and receiving financial resources in the system.

2.1.9.1 Resolved and Unresolved FRI

An FRI can refer to a single financial resource, or multiple resources. If the FRI refers to multiple resources, such as an account holders default accounts, the currency of the transaction identifies the exact financial resource. The FRI, for example FRI:46733495000/MSISDN, for an account holder with SEK and EUR accounts, can be used to address either of the accounts, depending on the currency of the transaction.

A resolved FRI always refers to a single resource, and an unresolved FRI might refer to multiple resources.
An unresolved FRI will be transformed into a resolved FRI using the currency of the transaction.
The following operations do not have a currency in the request and therefore have limited multi-currency support. When used in a multi-currency context, it is required to use Resolved FRIs that point to a specific account (for example MM). If a non-resolved FRI (for example MSISDN) is used, the operation returns AMBIGUOUS_CURRENCY.

A valid FRI matches the pattern FRI:[datavalue]/[type], where [datavalue] consists of non-whitespace characters, and [type] is a system predefined type.

The following table describes the defined FRI types:

FRI Types				
Type	Resolved or Unresolved	Points to	Description	Format
ALIAS	U	Default accounts	An alias to a user in the system.	`FRI:<alias>/ALIAS` or `FRI:<alias>@<subtype>/ALIAS`
BANK	R	External account	Identifies the bank.	`FRI:<bankaccountnumber>@<bankcode>/BANK`
BANK	R	External account	Identifies the bank.	`FRI:<bankcode>-<bankaccountnumber>/BANK`
CA	R	Specific account	Identifies a transfer between custody accounts.	`FRI:/CA`
CARD	U	Specific account	The `<card reference number>` is the Id of the card in the card processor system and is set by the card processor.	`FRI:<card reference number>@<identity>/CARD`

			CARD is used to identify a resource at a card processor. The `<identity>` is the unique username of the card processor in Ericsson Wallet Platform. The `<identity>` is set when the card processor is created and activated, see section **Card Management** in System Administrators Guide . See also Section 56.1.4.	
CASHIER	U	Default account of the employee.	Identifies the transferring Point of Sale (POS) employee at a specific merchant.	FRI:*<employee-id>*@*<merchant-alias>*/CASHIER
CASHVOUCHER	R	Cash voucher	Identifies a transfer to a cash voucher	FRI:*<cashvoucherid>*/CASHVOUCHER
COUPON	R	Coupon	Identifies the sending account when a coupon is redeemed.	FRI:*<coupon-id>*/COUPON
EMAIL	U	Default accounts	An email address of a user in the system.	FRI:*<local-part>*@*<domain>*/EMAIL
EPI-CARD	R	External Payment Instrument	Identifies a card which is registered as an External Payment Instrument.	FRI:*<external payment instrument ID>*/EPI-CARD
EXT	U	Default accounts	The external identifier for a user in the system.	FRI:*<external ID>*/EXT
FRIALIAS	U	Specific account or accounts	To use this type of FRI for transfers or payments an alias to a real FRI must be created. `AddFriAlias` operation is used to create an alias to a real FRI.	FRI:*<alias>*/FRIALIAS
ID	U	Default accounts	The internal account holder ID.	FRI:*<internal identifier>*/ID
INVITATION	R	Invitation	Points to the invitation account that is used when an invitation is created	FRI:*<invitation-token>*/INVITATION:
LOY	R	Default loyalty account	Point to the loyalty account of the current user, has no data value.	FRI:/LOY
MIGRATION	R	Old system account	Gives Ericsson Wallet Platform a reference to the account in the old system. Only for migration, not to be used for any operation	FRI:*<old-account-number>*/MIGRATION
MM	R	Specific account	An ID of an internal account in a mobile money system.	FRI:*<account number>*/MM
MPOS	U	Default account of the employee.	Identifies the transferring Point of Sale (POS) employee at a specific POS MSISDN.	FRI:*<employee-id>*@*<pos-msisdn>*/MPOS
MSISDN	U	Default accounts	Subscriber's MSISDN.	FRI:*<misdn>*/MSISDN

			The MSISDN shall be specified in international format as defined in Reference [16].	
POS	U	Default account of the employee that currently has the POS acquired.	Identifies the transferring Point of Sale (POS) device connected to a specific merchant.	FRI:*<pos-name>*@*<merchant-alias>*/POS
SAVINGS	U	Savings account	Identifies a transfer to a savings account.	FRI:*<savings-account-id>*/SAVINGS
SP	U	External accounts	The *<customer_ID>* is the customer's ID in the service provider system and is set by the service provider. SP is used to identify a resource at a service provider. The *<identity>* is the unique username of the service provider in Ericsson Wallet Platform. The *<identity>* is set when the service provider is created and activated, see section **Service Provider Management** in System Administrators Guide . See also Section 56.1.3.	FRI:*<customer_ID>*@*<identity>*/SP
USER	U	Default accounts	A username of a user in the system.	FRI:*<username>*/USER
VOUCHER	R	Voucher	Identifies a transfer to a voucher.	FRI:*<voucherid>*/VOUCHER
X- *<type>*	R	External account	Identifies a linked account at an account provider. *<type>* is configurable in externalaccount-services.xml	FRI:*<account number>*/X-*<type>*

2.1.9.2 ID

A valid identity associated with the account holder. A valid identity should match the pattern ID:[datavalue]/[type], where [datavalue] consists of non-whitespace characters, and [type] is predefined constant. The [type] could, for example be MSISDN, USER, EMAIL, or EXT, see table below for available types.

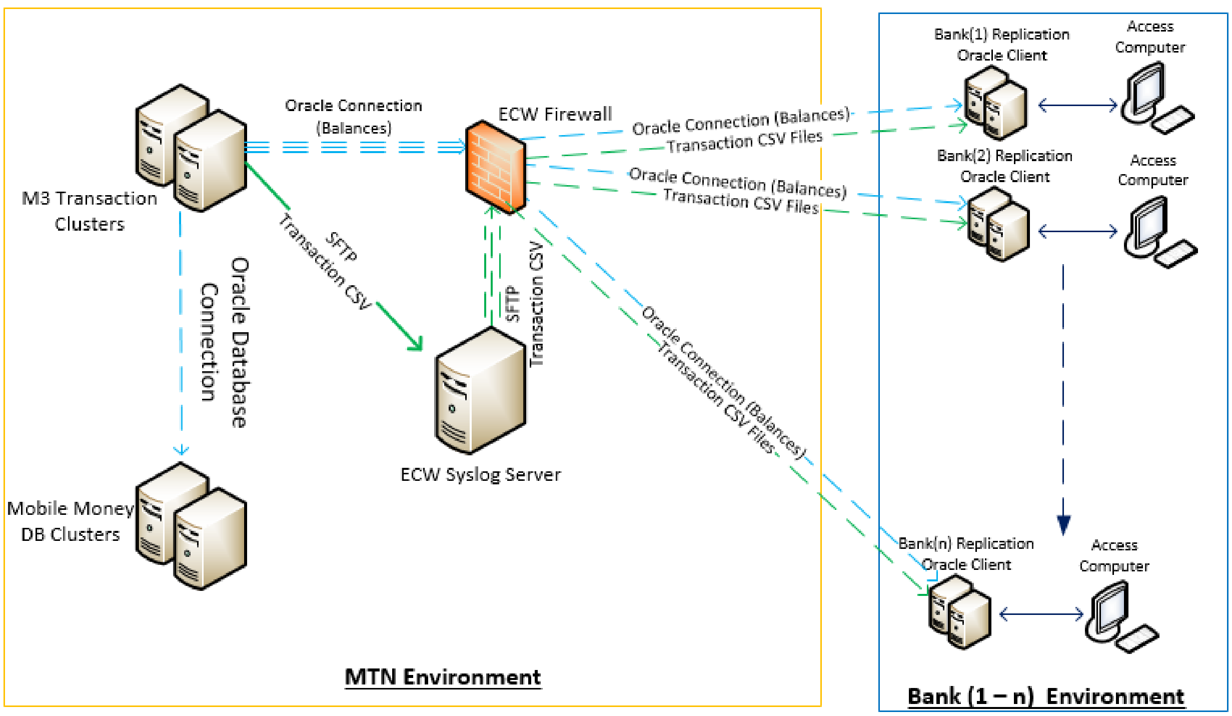
Table 1855 ID Types

Type	Description	Format
ALIAS	An alias to a user in the system.	ID:*<alias>*/ALIAS or ID:*<alias>*@*<subtype>*/ALIAS

EMAIL	An email address of a user in the system.	ID:<local-part>@<domain>/EMAIL
EXT	The external identifier for a user in the system.	ID:<external ID>/EXT
ID	The internal account holder ID.	ID:<internal identifier>/ID
MSISDN	The subscriber's MSISDN.	ID:<msisdn>/MSISDN
MPOS	The ID of an employee connected to a POS MSISDN.	ID:<employee-id>@<pos-msisdn>/MPOS
USER	A username of a user in the system.	ID:<username>/USER

2.1.10 Regulatory Compliance flow

From a compliance view, all transactions in the mobile money platform must be replicated to an escrow partner, a server deployed in the bank environment will be updated with balances but they bank will only have view access to subscriber data from the same. Data is replicated in real time into an oracle environment in the bank.



2.1.11 Prerequisites

The bank must be able to handle duplicate transactions and can process transactions on their end with the correct TLS version.

2.1.12 Pre-Conditions

The user should be in activated state.

2.2 Mobile Money Integration into MTN

2.2.1 Connectivity

The below URLs will be used for integration into the Mobile Money system, this will be done through a VPN.

Inbound traffic: The URI will depend on the API and is similar to the API request, e.g getbalancerequest the URI becomes getbalance, debitrequest becomes debit e.t.c

Test Platform:

https://10.156.145.219:<port>/bank/<uri

Production Primary Side:

https://10.156.144.21:<port>/bank /<uri>

Production Geo Side:
https://10.156.191.21:<port>/bank<uri>

Outbound traffic: All traffic from the mobile money platform to the 3pp will also have a URI appended at the end of the URL, if a 3pp has a URL like https://3ppendpint.com/ and we are sending a payment or getfinancialresource information request to them, the final end point will be https://3ppendpoint.com/deposit or https://3ppendpoint.com/paymentinstruction

2.2.2 Client authentication

The following authentication methods are supported:
Presentation layer - server side certificate

Authentication of the Mobile Money Manager service is supported through an HTTPS server certificate. In order to successfully connect to the interface, this certificate has to be trusted by the client, either directly, or through a certificate trust chain.

No matter if server side certificates are used or not, the Application (the Service Provider system or the Service Bus) still needs to authenticate itself with a user name and a password. There are two methods supported in the Mobile Money Manager for this.

2.2.2.1 Application layer – normal authentication

Authentication of clients of the service is performed by a user name and password-based login service, described in 2.3.1.1.1

The system returns the following in the HTTP header for the response:

Set-Cookie: sessionid=default8fcee064690b45faa9f8f6c7e21c5e5a

The session ID is then used as a cookie in the HTTP header in the following requests towards the Mobile Money Manager.

Cookie: sessionid=default8fcee064690b45faa9f8f6c7e21c5e5a

2.2.2.1.1 Login Service

A successful login to the system creates a session ID. The session ID can be used for multiple operations and will be kept alive and have its lifetime extended as long as operations are performed within the session lifetime.

Identifying oneself to the system by entering one's credential.

loginrequest:

loginrequest			
Name	Type	Multiplicity	Description
identity	String	Mandatory	Identity of user.
credential	credential information	Mandatory	Credential used to verify the users identity.

loginresponse:
An empty response. If HTTP response code 200 is returned the login is successful. If HTTP response code 302 is returned either a secondarylogin with an OTP or a changepassword are expected is needed before successful login. Which one is indicated in the response header.

2.2.2.1.2 credential information

credential			
Name	Type	Multiplicity	Description
secret	string	Mandatory	Credential secret
type	string	Mandatory	Credential secret format. Must be set to `string`.

2.2.2.2 Application layer - basic authentication

The protocol also supports basic authentication scheme as defined in RFC 1945. This can be used instead of session establishment for performing an operation without requiring login.

The credentials need to be passed in the HTTP header in the following format:

Authorization: Basic QWxhZGRpbjpvcGVuIHNlc2FtZQ==

Where the credentials are base64 encoded and in the format <login>:<password>.

2.2.3 Protocol Structure

This section describes the structure of the protocol.

2.2.3.1 URI Syntax

The URI of the HTTP request is structured according to the following pattern:
URI := /{context}/{method}
where
context - is a fixed string representing the routing entity e.g. /poextvip

method is a fixed string representing the operation type like debit,payment,sptransfer e.t.c.

The same method can be used by different users. Depending on the user's authentication, a different context is used.

2.2.3.2 Response Status Codes

The protocol uses the regular HTTP status codes/ Status codes in the 200 range imply that the request completed normally. In this case the response body will be formatted according to the expected response syntax for the specified method. Status code in the 300 range indicate that the request must be resent to a different location. Status codes in the 400 or 500 ranges imply that there was an error executing the request.

Used HTTP response codes

Response Code	Description
200 (OK)	Sent when the request completed successfully.
202 (Accepted)	The request has been accepted for processing, but the processing has not been completed (pending).
302 (Moved Temporarily)	The URL provided in the Location header field indicates where to send the following request and which request type to use. The following request types are possible:
401 (Unauthorized)	The request needs authentication.
403 (Forbidden)	The request was denied, and will be denied also in the future.
404 (Not found)	The resource specified in the URI was not found.
500 (Internal Server Error)	The server encountered a general error during execution, see Section 57.

2.2.4 Message Integrity Protection

This section describes how the message integrity is protected by message signatures.
Message signing is implemented so that a random challenge, a timestamp and selected values from the request or response messages, are included in the signature. The signature string is then delivered together with the message in a HTTP request or response header named X-Signature

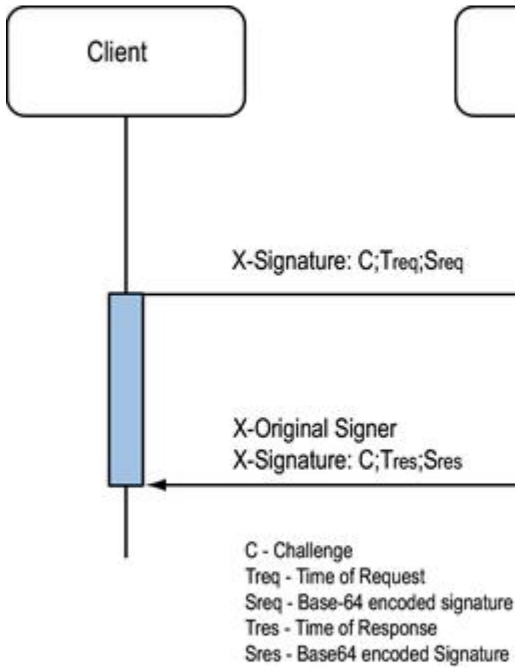


Fig xx: Request and Response Signing

The signature header string consists of the three following parts, separated by semicolon:

- **random-challenge**, a random string chosen by the client, unique during the message life-time. The server will reject messages with the same random-challenge.
- **timestamp**, the number of seconds since 1 Jan 1970, 00:00:00 UTC (also known as **POSIX time**).
- **b64-encoded-signature**, the Base64 encoded signature.

The signature is created by using the SHA256 digest together with an RSA encryption over the UTF-8 encoding of the to-be-signed string.

The to-be-signed string is composed of the three following parts, separated by a semicolon.

- **random-challenge**, the same random challenge string as in the signature header string.
- **timestamp**, the same time stamp string as for the signature header string.

values-to-be-signed, are made up of zero or more values from the request or response message elements. The values are specific to each message, and are listed in each section in this protocol specification.

For cases when no values are to be part of the signature, the to-be-signed string is composed as:

random-challenge;timestamp;

Note:

The string ends with a semicolon.

If message signing is enabled on an incoming request context or outgoing client, all messages must be signed. That also applies for the messages that do not contain any message element values to be included in the signature.

2.2.4.1 Message Signing

This section describes an example flow of the message signing procedure.

- Below is an example of a request, and its associated response:

```
<request>
    <transactionid>123456</transactionid>
    <providertransactionid>AREF2014015</providertransactionid>
    <newbalance>
        <amount>123.00</amount>
        <currency>EUR</currency>
    </newbalance>
    <paymenttoken>tokenstring1</paymenttoken>
    <message>Thanks for all the fish</message>
    <status>COMPLETED</status>
</request>
```

- The client sending the request chooses the data to be used:

```
random-challenge: 2siqin0aj4kbsisiqj7cs8s4u5
timestamp-request: 1402997614
```

The request requires the following values to be signed, transactionid, providertransactionid, status, the to-be-signed string will look like the following:

2siqin0aj4kbsisiqj7cs8s4u5;1402997614;123456;AREF2014015;COMPLETED

This is converted to bytes, using UTF-8 encoding, and signed with the clients private key, using SHA-256 as hash algorithm. The resulting bytes are then converted to Base-64 encoding, for example:

```
b64-encoded-signature: ZHNmZ25ma2dramdsbHNkZnNrZnllZ2hmamZnaGRnbGdkaGdm
```

- The resulting HTTP request header will look like the following:

```
X-Signature: 2siqin0aj4kbsisiqj7cs8s4u5;1402997614;\
ZHNmZ25ma2dramdsbHNkZnNrZnllZ2hmamZnaGRnbGdkaGdm
```

- When the server responds to the request, it sends the following response:

```
<response>
</response>
```

- The response signature is calculated like below:

```
random-challenge: 2siqin0aj4kbsisiqj7cs8s4u5              (same value as
in the request)
timestamp-response: 1402997615                  (servers own timestamp)
```

- As this response contains no parameter values to be signed, the to-be-signed string will look like:

  `2siqin0aj4kbsisiqj7cs8s4u5;1402997615;`

- The to-be-signed string is converted to bytes, using UTF-8 encoding, and signed with the servers private key, using SHA-256 as hash algorithm. The resulting bytes are then converted to Base-64 encoding, for example:

```
b64-encoded-signature: dzQ1Njd1M2tsajkwNHRlcnR0NDIzMDllZQ==
```

- The resulting response header looks like:

```
        X-Signature:
2siqin0aj4kbsisiqj7cs8s4u5;1402997615;dzQ1Njd1M2tsajkwNHRlcnR0NDIzMDllZQ
==
```

2.2.4.2 Original Signer Header

A system responding to a signed client request from the Mobile Money Manager have to reply with a signed response.

In order to indicate which system it is, the response must also contain another HTTP header, called X-Original-Signer.

The value of this header is the identity of the system, for example ID:<bankusername>/USER. This value is then used by the Mobile Money Manager to look up the corresponding public key for the account holder, and verify the response signature.

2.2.5 SSL Certificate setup

Follow the following steps for TLS security setup, this is a guideline is used to ensure we have TLS setup, we use the Mobile Money certificate authority to sign the certificates of the entity integrating to the Mobile Money Platform.

- A 3rd party creates a TLS certificate signing request (CSR) that is to be signed by the Mobile Money Platform
- The Mobile Money Platform signs that TLS CSR with Mobile Money Certificate Authority and share CSR response certificate as well as CA file with 3pp
- 3pp needs to import that certificate and CA in their keystore as trusted ones
- TLS v1.2 should be used
- On the mobile money platform, mutual authentication will be enabled to ensure only trusted entities are allowed

** Certificate renewal tracking is the responsibility of the 3pp

2.2.5.1 CSR creation guidelines for Mobile Money

- Other security requirements for TLS.
- Signature Algorithm must be SHA256 or above
- Private keys MUST be created and stored within the service host i.e. Importing of Private keys is NOT Allowed.
- Only public keys or certificates can be Exported, Private keys Must not be Exported.

- All certificates MUST be issued for a specific service or host (e.g. ewp.ug.com or mobileapp.ug.com) and SHOULD NOT contain any wild cards.
- All Subject Fields must be fully filled, Mandatory Fields CN,OU, O,L,ST,C and E
- All connection (incoming & outgoing) must terminate at the EWP firewall (F5) to enable traffic inspection.
- Deny all traffic in EWP Firewall by default (Blacklist) and traffic shall be permitted only on a need basis (Whitelist).
- All connections must use TLS1.2 and above
- All 3PP and operator's IT systems must use TLS mutual authentication
- TLS encryption keys must follow minimum requirements for key generation (e.g. key size of at least 2048 bits)

3 Exclusion

Any code implementation and validation of bank end code validation is not included in this document

4 Assumptions

- The existing Architecture in ECW will not change.
- The SSL VPN will still be maintained.
- The only change required from ECW is to return a Response Code Reference for every menu Item that needs redirect.
- All 3rd Party interaction will be done on the 3PP menu system.
- 3PP will then send a debit request to ECW.
- Subscriber will use USSD Push or dial the Mobile Money menu to approve the debit.

5 Impact

This section covers the impact on existing functionality

6 Notifications

6.1 SMS Notifications

Existing SMSs apply

6.2 USSD Notifications

6.3 XML Notifications

7 References

[1] URS

7.1 Glossary

3PP = 3rd Party Product

ECW = Ericsson Converged Wallet (Also the mobile money system)

PO = Partner Orchestrator (used as a gateway into Mobile Money)

XML = Extensible Mark Up Language

USSD = Unstructured Supplementary Data

SMS = Short Messaging Protocol

SSL = Secure Socket Layer

8 Approval

This document has been mutually reviewed and approved to constitute the full scope of the solution to be delivered by the MTN Uganda and the bank for escrow liquidation. This document is signed in two (3) copies of which the parties have taken one (1) each.

MTN Uganda:

Name	Role	Signature	Date

Ericsson:

Name	Role	Signature	Date

Bank:

Name	Role	Signature	Date

9 Appendix

9.1 Reporting logs;

JSON LOGS (TRANSFER FROM CC-GUI)

- {"Header":{"Type":"TXN","Ver":"1.0"},"Instruction":{"Header":{"Time":"2017-08-14T21:41:49.450+0300","Tid":11402922,"Fid":2651174,"Uid":3,"Sid":1500007633539,"User":"ID:ccuser1/ADMIN","RealUser":"ID:ccuser1/ADMIN"}},"Status":{"Code":"CREATED"}}

- {"Header":{"Type":"RESERVATION","Ver":"1.0"},"Instruction":{"Header":{"Type":"TRANSFER_TO_ANY_BANK_ACCOUNT","Time":"2017-08-14T21:41:49.450+0300","Tid":11402922,"Fid":2651174,"Uid":4,"Sid":1500007633539,"User":"ID:ccuser1/ADMIN","RealUser":"ID:ccuser1/ADMIN"},"Amount":2000.00,"Cc":"UGX","From":{"Fri":"FRI:1300756/MM","RFri":"FRI:1300756/MM","FROwner":"ID:320668/ID","SP":"Internal","OffNet":false,"Rate":1,"Cc":"UGX","Amount":2000.00,"Committed":{"Before":10129687.00,"After":10127687.00},"Available":{"Before":10122687.00,"After":10120687.00},"Total":{"Before":10129687.00,"After":10127687.00},"BankDomainName":"MTNUGANDA"},"To":{"Fri":"FRI:1234134134134@12345/BANK","RFri":"FRI:1000020/MM","SP":"Internal","OffNet":false,"Rate":1,"Cc":"UGX","Amount":2000.00,"BankDomainName":"MTNUGANDA"}},"Status":{"Code":"EXECUTED"}}

- {"Header":{"Type":"TXN","Ver":"1.0"},"Instruction":{"Header":{"Time":"2017-08-14T21:41:50.175+0300","Tid":11402923,"Fid":2651174,"Uid":5,"Sid":1500007633521,"User":"ID:FinancialTransactionService/SERVICE","RealUser":"ID:FinancialTransactionService/SERVICE"}},"Status":{"Code":"COMMITTED"}}

JSON LOGS (TRANSFER FROM PARTNER PORTAL)

- {"Header":{"Type":"TXN","Ver":"1.0"},"Instruction":{"Header":{"Time":"2017-08-14T21:48:56.544+0300","Tid":11403119,"Fid":2651180,"Uid":16,"Sid":1500007633555,"User":"ID:320668/MM","RealUser":"ID:320668/MM"}},"Status":{"Code":"CREATED"}}

- {"Header":{"Type":"RESERVATION","Ver":"1.0"},"Instruction":{"Header":{"Type":"WITHDRAWAL","Time":"2017-08-14T21:48:56.544+0300","Tid":11403119,"Fid":2651180,"Uid":17,"Sid":1500007633555,"User":"ID:320668/MM","RealUser":"ID:320668/MM"},"Amount":2000.00,"Cc":"UGX","From":{"Fri":"FRI:1300756/MM","RFri":"FRI:1300756/MM","FROwner":"ID:320668/ID","SP":"Internal","OffNet":false,"Rate":1,"Cc":"UGX","Amount":2000.00,"Committed":{"Before":10127687.00,"After":10125687.00},"Available":{"Before":10120687.00,"After":10118687.00},"Total":{"Before":10127687.00,"After":10125687.00},"BankDomainName":"MTNUGANDA"},"To":{"Fri":"FRI:90000121314212@12345/BANK","RFri":"FRI:1000020/MM","SP":"Internal","OffNet":false,"Rate":1,"Cc":"UGX","Amount":2000.00,"BankDomainName":"MTNUGANDA"}},"Status":{"Code":"EXECUTED"}}

- {"Header":{"Type":"TXN","Ver":"1.0"},"Instruction":{"Header":{"Time":"2017-08-14T21:48:57.165+0300","Tid":11403123,"Fid":2651180,"Uid":21,"Sid":1500007633521,"User":"ID:FinancialTransactionService/SERVICE","RealUser":"ID:FinancialTransactionService/SERVICE"}},"Status":{"Code":"COMMITTED"}}

- **<u>From transaction csv;</u>**
- "4.0","11402922","2651174","","2017-08-14 21:41:50","ID:ccuser1/ADMIN","ID:ccuser1/ADMIN","320668","256772712561","Agaba","MTNU Mobile Money Subscriber No Rating","1300756","MM","0.00","0","0","MTNUGANDA","","","","","1000020","MM","0.00","0","0","MTNUGANDA","TRANSFER_TO_ANY_BANK_ACCOUNT","2000.00","UGX","COMMITTED","http-cc","",""
- "4.0","11403119","2651180","","2017-08-14 21:48:57","ID:320668/MM","ID:320668/MM","320668","256772712561","Agaba","MTNU Mobile Money Subscriber No Rating","1300756","MM","0.00","0","0","MTNUGANDA","","","","","1000020","MM","0.00","0","0","MTNUGANDA","WITHDRAWAL","2000.00","UGX","COMMITTED","http-partner","",""

9.2 API additional parameters

9.2.1 extensiontype v1.0

Wrapper type holding external systems extra data in a list of XML element nodes. A unique namespace has to be used for this type.

Parameters

Appendix Table 1 extensiontype

Name	Type	Optionality	Description
any	list<element>	Optional	Use to send any other information.

9.2.2 financialtransactionstatus v1.0

The status of a financial transaction.

Parameters

Appendix Table 2 Enumeration

Value	Description
FAILED	Failed transaction.
PENDING	Pending transaction.
SUCCESSFUL	Successful transaction.

9.2.3 loyalty v1.0

Contains information about loyalty.

Appendix Table 3 loyalty

Name	Type	Optionality	Description
generated	decimal	Optional	The rewarded loyalty, generated amount.
consumed	decimal	Optional	The loyalty fee, consumed amount.
newbalance	decimal	Optional	The loyalty balance after execution.

9.2.4 loyaltypointstype v1.0

Contains the amount of loyalty points.

Appendix Table 4 loyaltypointstype

Name	Type	Optionality	Description
amount	decimal	Mandatory	The amount of loyalty points

9.2.5 moneydetailstype v1.0

Contains the amount and currency of a monetary value.

Appendix Table 5 moneydetailstype

Name	Type	Optionality	Description
amount	decimal	Mandatory	The monetary amount. Validated with IsAmount (allowNull=false) (allowZero=true).
currency	string	Mandatory	The currency, as an ISO 4217 formatted string. Validated with IsCurrencyCode. Parameter can not be NULL.

9.2.6 deposit v1.0

Sent by the bank to the system with detailed information about the deposit that should be performed. This operation supports reservation

```
URI: /{baseContext}/deposit
Namespace: http://www.ericsson.com/em/emm/settlement/v1_0
```

depositrequest:

Appendix Table 6 depositrequest

Name	Type	Optionality	Description
bankcode	string	Optional	Custody account bank code. This along with custody account number may be used to identify the custody account. Validated with IsBankCode.
accountnumber	string	Mandatory	Custody account number. This along with custody bank code will be used to identify the custody account. Parameter can not be NULL. Validated with IsBankAccount.
transactiontimestamp	datetime v1.0	Mandatory	Date and time when the transaction was initiated. Parameter can not be NULL. Validated with IsDateTimeValue.
amount	moneydetailstype v1.0	Mandatory	Amount to be deposited to the receiver. Parameter can not be NULL. Validated with IsPositiveAmount.
receiver	string	Mandatory	A valid FRI referring to the receiver of the deposit. Parameter can not be NULL. Validated with Length (max=100). Validated with IsFRI.
banktransactionid	string	Mandatory	The bank's transaction ID. Validated with NotBlank. Validated with Length (max=100). Validated with IsExternalReferenceString.
receiverfirstname	string	Optional	The receiver's first name. Validated with IsFirstname.
receiversurname	string	Optional	The receiver's surname. Validated with IsSurname.
message	string	Optional	Optional message which will appear in the receiver's transaction history. Validated with Length (max=100). Validated with IsRestrictedString.

Values to be signed: `bankcode`, `accountnumber`, `amount/amount`, `amount/currency`, `receiver`, `banktransactionid`

depositresponse:

Appendix Table 7 depositresponse

Name	Type	Optionality	Description
status	depositstatus v2.0	Mandatory	The status of the deposit.
financialtransactionid	string	Optional	Contains the financial transaction ID. Validated with Length (max=40). Validated with IsPositiveLong (allowZero=false).
failurereason	string	Optional	Contains the failure reason if status is SUSPENDED. Validated with Length (max=200). Validated with IsRestrictedString.

Values to be signed: `status`, `financialtransactionid`

9.2.7 paymentinstruction v2.0

Sends a payment instruction request to a bank with details about the transaction that should be performed.

URI: /{baseContext}/paymentinstruction
Namespace: http://www.ericsson.com/em/emm/settlement/v2_0

paymentinstructionrequest:

Appendix Table 8 paymentinstructionrequest

Name	Type	Optionality	Description
transactiontimestamp	datetime v1.0	Mandatory	Date and time. Validated with IsDateTimeValue. Parameter can not be NULL.
amount	moneydetailstype v1.0	Mandatory	The amount that should be withdrawn. Parameter can not be NULL. Validated with IsPositiveAmount.
paymentinstructionid	string	Mandatory	Payment instruction ID. Validated with NotBlank. Validated with Length (max=32). Validated with IsLongString (allowNegativeValue=false).
receiverbankcode	string	Optional	The receiver's bank code. Validated with IsBankCode.
receiveraccountnumber	string	Mandatory	The receiver's account number. Parameter can not be NULL. Validated with IsBankAccount.
receiverfirstname	string	Optional	The receiver's first name. Validated with IsFirstname.
receiversurname	string	Optional	The receiver's surname. Validated with IsSurname.
message	string	Optional	Optional message which will appear in the receiver's transaction history. Validated with Length (max=100). Validated with IsRestrictedString.
receivermsisdn	string	Optional	The msisdn of the receiving party. Validated with IsMSISDN.
transmissioncounter	string	Mandatory	A counter representing the number of times a particular request has been transmitted. The counter is set to 1 the first time the request is transmitted and is incremented by 1 for each retransmission of the request. Validated with NotBlank. Validated with IsIntegerStringWithinRange (min=0).
transactionid	string	Mandatory	A unique identifier for the financial transaction. Validated with NotBlank. Validated with Length (max=32). Validated with IsLongString (allowNegativeValue=false).

Values to be signed: `amount/amount`, `amount/currency`, `paymentinstructionid`, `receiverbankcode`, `receiveraccountnumber`

paymentinstructionresponse:

Appendix Table 9 paymentinstructionresponse

Name	Type	Optionality	Description
status	paymentinstructionresult v2.0	Mandatory	Status of the payment instruction from the bank. Parameter can not be NULL.
paymentinstructionid	string	Mandatory	Payment instruction ID as received from EWP. Validated with NotBlank. Validated with Length (max=32). Validated with IsLongString (allowNegativeValue=false).
banktransactionid	string	Mandatory	The bank's transaction ID. Validated with NotBlank. Validated with Length (max=100). Validated with IsExternalReferenceString.
transactiontimestamp	datetime v1.0	Mandatory	Date and time as sent in the payment instruction request. Parameter can not be NULL. Validated with IsDateTimeValue.
bookingtimestamp	datetime v1.0	Mandatory	Date and time when the bank made the withdrawal. Parameter can not be NULL. Validated with IsDateTimeValue.
amount	moneydetailstype v1.0	Mandatory	The amount that has been withdrawn. Parameter can not be NULL. Validated with IsPositiveAmount.

Values to be signed: `status`, `paymentinstructionid`, `banktransactionid`, `amount/amount`, `amount/currency`

9.2.8 validateaccountholder v1.0

Used by the service provider to check if a certain identity is a valid account holder in Mobile Money Manager.

`URI: /{baseContext}/validateaccountholder`

validateaccountholderrequest:

Appendix Table 10 validateaccountholderrequest

Name	Type	Optionality	Description
accountholderid	string	Mandatory	The identity of the account holder to validate. Example: ID:46733491234/MSISDN. Parameter can not be NULL. Validated with IsIdentity.

validateaccountholderresponse:

Table 62 validateaccountholderresponse

Name	Type	Optionality	Description
valid	boolean	Mandatory	TRUE means that requested identity is an account holder that can receive any transaction. FALSE: Means that identity either not found or it

			is not able to receive any transaction. Parameter can not be NULL.

9.2.9 getaccountholderpersonalinformation v1.0

Returns personal information connected to the identity sent in the request.

URI: /{baseContext}/getaccountholderpersonalinformation

Namespace: http://www.ericsson.com/em/emm/provisioning/v1_0

getaccountholderpersonalinformationrequest:

Appendix Table 11 getaccountholderpersonalinformationrequest

Name	Type	Optionality	Description
identity	string	Optional	A valid identity associated with the account holder. Validated with IsIdentity.

getaccountholderpersonalinformationresponse:

Appendix Table 12 getaccountholderpersonalinformationresponse

Name	Type	Optionality	Description
information	personalinformation v1.0	Optional	The account holder's personal information.

9.2.10 getcustodyaccounttransactionhistory v1.3

URI: /{baseContext}/getcustodyaccounttransactionhistory

Namespace: http://www.ericsson.com/em/emm/financial/v1_3

getcustodyaccounttransactionhistoryrequest:

Appendix Table 13 getcustodyaccounttransactionhistoryrequest

Name	Type	Optionality	Description
bankcode	string	Optional	Custody account bank code. Validated with IsBankCode.
accountnumber	string	Mandatory	Custody account number. Parameter can not be NULL. Validated with IsBankAccount.
numoftransactions	integer	Mandatory	The number of transactions to return in the response. Default value: 50. Minimum value: 1. Parameter can not be NULL.
indexoffset	integer	Optional	Start retrieving transaction history from this index.
financialtransactionid	long	Optional	Retrieve transaction history with this financial transaction ID. Validated with IsPositiveLong (allowZero=false).

externaltransactionid	string	Optional	Retrieve transaction history with this external transaction ID. Validated with IsExternalReferenceString.
transactionstatus	financialtransactionstatus v1.0	Optional	Retrieve transaction history with this status.
transactiontype	list< financialinstructiontype v2.1>	Optional	Retrieve transaction history with this transaction type. It is possible to have more than one transactiontype element to filter on. Shows the financial transaction type. Validated with NoNullElements.
startdatefrom	date	Optional	Retrieve transaction history created from this start date. Validated with IsDateTime.
startdateto	date	Optional	Retrieve transaction history created until this stop date. Validated with IsDateTime.
datefrom	date	Optional	Retrieve transaction history finalized (SUCCESSFUL, FAILED) from this start date. Validated with IsDateTime.
dateto	date	Optional	Retrieve transaction history finalized (SUCCESSFUL, FAILED) until this stop date. Validated with IsDateTime.
direction	direction v1.0	Optional	Retrieve transaction history with the specified direction, that is, incoming or outgoing transfers.
otherfri	string	Optional	The FRI of the other party in transfer, could be from or to depending on direction. Validated with IsFRI.
quoteid	quoteid v1.0	Optional	The ID of the quote to use.

getcustodyaccounttransactionhistoryresponse:

Appendix Table 14 getcustodyaccounttransactionhistoryresponse

Name	Type	Optionality	Description
totalcount	integer	Optional	Total amount of custody account transaction history entries available with the specified condition.
entry	list< getcustodyaccounttransactionhistoryentrytype v1.3>	Optional	Contains a list of transactions, each transaction contains type, amount, date and so on. Contains the main summary of a custody account transaction history event. Validated with NoNullElements.
loyalty	loyalty v1.0	Optional	Loyalty information generated with the transaction.